3 Yr. Degree/4 Yr. Honours 3rd Semester Examination, 2024 (CCFUP)

Subject : Computer Science

Course: COMP3012 (MAJOR)

(Data Structures)

Time: 2 Hours **Full Marks: 40**

The figures in the right hand margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* of the following questions: 2×5=10

(a) Define a stack and mention two of its primary operations.

(b) What will be the output of the following stack operations?

Push (10), Push (20), Pop (), Push (30), Pop (), Pop ().

(c) What is the purpose of a "head" pointer in a linked list?

(d) Differentiate between linear queue and circular queue.

(e) What is the difference between a full binary tree and a complete binary tree?

(f) Explain the term 'depth' of a node in a tree.

(g) What is Abstract Data Type (ADT)?

(h) What is the best-case time complexity of Bubble Sort and when does it occur?

2. Answer *any two* of the following questions: 5×2=10

(a) Write an algorithm to reverse a singly linked list. Explain how pointers are used during the reversal process.

(b) Show the step-by-step evaluation of the postfix expression 62/3-42*+ using a stack. Differentiate between Stack and Queue. 3+2

(c) Differentiate between open addressing and chaining as collision resolution techniques in hashing. Provide an example of chaining.

(d) Write an algorithm to sort an array using insertion sort.

3. Answer *any two* of the following questions: 10×2=20

(a) How does binary search compare to linear search in terms of time complexity? Write the algorithm for binary search and explain its implementation with an example. 2+3+5

(b) What is Binary Search Tree (BST)? Write an algorithm to insert a node into a Binary Search Tree. Also, explain how to search for a node in a BST. 2+5+3

(c) What is a doubly linked list? Discuss its advantages and disadvantages compared to a singly linked list. Implement a function to delete a node from the end of a doubly linked list. 2+3+5

(d) Write short notes on *any two*: 5×2

(i) Circular Queue

(ii) Recurdsion

(iii) Hash function

(iii) Sparse Matrix